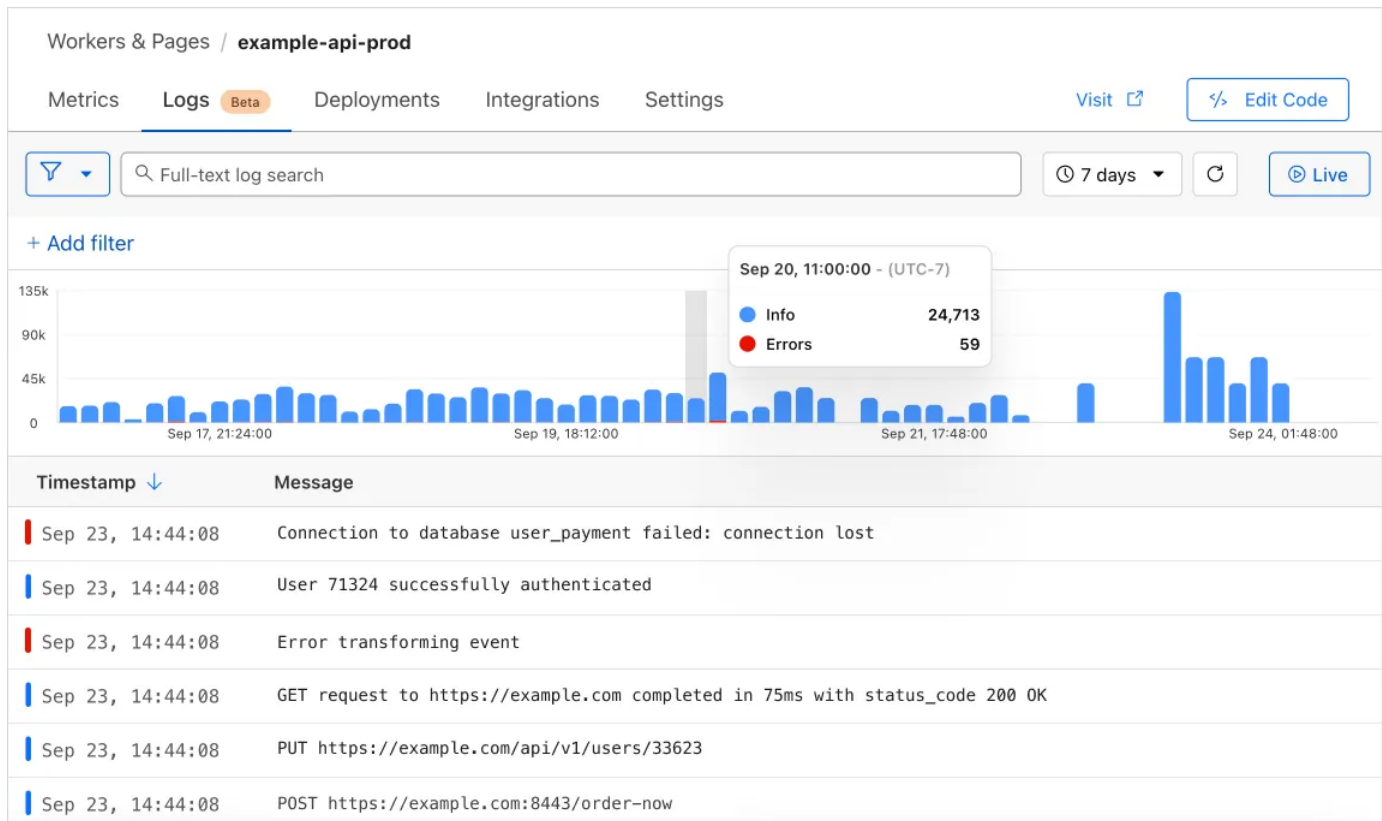


Workers Logs

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Workers Logs lets you automatically collect, store, filter, and analyze logging data emitted from Cloudflare Workers. Data is written to your Cloudflare Account, and you can query it in the dashboard for each of your Workers. All newly created Workers will come with the observability setting enabled by default.

Logs include [invocation logs](#), [custom logs](#), errors, and uncaught exceptions.



To send logs to a third party, use [Workers Logpush](#) or [Tail Workers](#).

Enable Workers Logs

Wrangler version

Minimum required Wrangler version: 3.78.6. Check your version by running `wrangler --version`. To update Wrangler, refer to [Install/Update Wrangler](#).

You must add the observability setting for your Worker to write logs to Workers Logs. Add the following setting to your Worker's Wrangler file and redeploy your Worker.

`{}` wrangler.jsonc

 wrangler.toml

```
{
  "observability": {
    "enabled": true,
    "head_sampling_rate": 1
  }
}
```

[Head-based sampling](#) allows you set the percentage of Workers requests that are logged.

Enabling with environments

[Environments](#) allow you to deploy the same Worker application with different configurations. For example, you may want to configure a different `head_sampling_rate` to staging and production. To configure observability for an environment named `staging`:

1. Add the following configuration below `[env.staging]`

 `wrangler.jsonc` `wrangler.toml`

```
{
  "env": {
    "staging": {
      "observability": {
        "enabled": true,
        "head_sampling_rate": 1
      }
    }
  }
}
```

2. Deploy your Worker with `npx wrangler deploy -e staging`
3. Repeat step 1 and 2 for each environment.

View logs from the dashboard

Access logs for your Worker from the [Cloudflare dashboard](#).

- 1 Log in to the [Cloudflare dashboard](#) and select your account.
- 2 In Account Home, go to **Workers & Pages**.
- 3 In **Overview**, select your **Worker**.
- 4 Select **Logs**.

Best Practices

Logging structured JSON objects

To get the most out of Workers Logs, it is recommended you log in JSON format. Workers Logs automatically extracts the fields and indexes them intelligently in the database. The benefit of this structured logging technique is in how it allows you to easily segment data across any dimension for fields with unlimited cardinality. Consider the following scenarios:

Scenario	Logging Code	Event Log (Partial)
1	<code>console.log("user_id: " + 123)</code>	<code>{message: "user_id: 123"}</code>
2	<code>console.log({user_id: 123})</code>	<code>{user_id: 123}</code>
3	<code>console.log({user_id: 123, user_email: "a@example.com"})</code>	<code>{user_id: 123, user_email: "a@example.com"}</code>

The difference between these examples is in how you index your logs to enable faster queries. In scenario 1, the `user_id` is embedded within a message. To find all logs relating to a particular `user_id`, you would have to run a text match. In scenarios 2 and 3, your logs can be filtered against the keys `user_id` and `user_email`.

Features

Invocation Logs

Each Workers invocation returns a single invocation log that contains details such as the Request, Response, and related metadata. These invocation logs can be identified by the field `$cloudflare.$metadata.type = "cf-worker-event"`. Each invocation log is enriched with information available to Cloudflare in the context of the invocation.

In the Workers Logs UI, logs are presented with a localized timestamp and a message. The message is dependent on the invocation handler. For example, Fetch requests will have a message describing the request method and the request URL, while cron events will be listed as cron. Below is a list of invocation handlers along with their invocation message.

Invocation logs can be disabled in wrangler by adding the `invocation_logs = false` configuration.

{} `wrangler.jsonc` `wrangler.toml`

```
{
  "observability": {
    "logs": {
      "invocation_logs": false
    }
  }
}
```

Invocation Handler	Invocation Message
Alarm	<Scheduled Time>
Email	<Email Recipient>
Fetch	<Method> <URL>
Queue	<Queue Name>
Cron	<UNIX-cron schedule>
Tail	tail
RPC	<RPC method>
WebSocket	<WebSocket Event Type>

Custom logs

By default a Worker will emit [invocation logs](#) containing details about the request, response and related metadata.

You can also add custom logs throughout your code. Any `console.log` statements within your Worker will be visible in Workers Logs. The following example demonstrates a custom `console.log` within a Worker request handler.

```
export default {
  async fetch(request) {
    const { cf } = request;
    const { city, country } = cf;

    console.log(`Request came from city: ${city} in country:
    ${country}`);

    return new Response("Hello worker!", {
      headers: { "content-type": "text/plain" },
    });
  },
};
```

After you deploy the code above, view your Worker's logs in [the dashboard](#) or with [real-time logs](#).

Head-based sampling

Head-based sampling allows you to log a percentage of incoming requests to your Cloudflare Worker. Especially for high-traffic applications, this helps reduce log volume and manage costs, while still providing meaningful insights into your application's performance. When you configure a head-based sampling rate, you can control the percentage of requests that get logged. All logs within the context of the request are collected.

To enable head-based sampling, set `head_sampling_rate` within the observability configuration. The valid range is from 0 to 1, where 0 indicates zero out of one hundred requests are logged, and 1 indicates every request is logged. If `head_sampling_rate` is unspecified, it is configured to a default value of 1 (100%). In the example below, `head_sampling_rate` is set to 0.01, which means one out of every one hundred requests is logged.

`{}` `wrangler.jsonc` `wrangler.toml`

```
{
  "observability": {
    "enabled": true,
    "head_sampling_rate": 0.01
  }
}
```

Limits

Description	Limit
Maximum log retention period	7 Days
Maximum logs per account per day ¹	5 Billion
Maximum log size ²	256 KB

¹ There is a daily limit of 5 billion logs per account per day. After the limit is exceeded, a 1% head-based sample will be applied for the remainder of the day.

² A single log has a maximum size limit of [256 KB](#). Logs exceeding that size will be truncated and the log's `$cloudflare.truncated` field will be set to true.

Pricing

Billing start date

Workers Logs billing will begin on April 21, 2025.

Workers Logs is included in both the Free and Paid [Workers plans](#).

Log Events Written	Retention
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	Log Events Written	Retention
Workers Free	200,000 per day	3 Days
Workers Paid	20 million included per month +\$0.60 per additional million	7 Days

Examples

Example 1

A Worker serves 15 million requests per month. Each request emits 1 invocation log and 1 `console.log`. `head_sampling_rate` is configured to 1.

	Monthly Costs	Formula
Logs	\$6.00	$((15,000,000 \text{ requests per month} * 2 \text{ logs per request} * 100\% \text{ sample}) - 20,000,000 \text{ included logs}) / 1,000,000 * \0.60
Total	\$6.00	

Example 2

A Worker serves 1 billion requests per month. Each request emits 1 invocation log and 1 `console.log`. `head_sampling_rate` is configured to 0.1.

	Monthly Costs	Formula
Logs	\$108.00	$((1,000,000,000 \text{ requests per month} * 2 \text{ logs per request} * 10\% \text{ sample}) - 20,000,000 \text{ included logs}) / 1,000,000 * \0.60
Total	\$108.00	

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Last updated: May 13, 2025

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